

Molecular Characteristics of Spontaneous Remission in Major Depressive Disorder: Changes in Serum Acetylcarnitine and Glycerophosphocholine Levels

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Objective: Spontaneous remission may influence the outcome of clinical trials and evaluation of antidepressant efficacy, as it is associated with placebo effects and false remission rates. However, the characteristics of spontaneous remission and its biological mechanisms remain poorly understood. This study aimed to explore the metabolic signatures and underlying biological mechanisms of spontaneous remission using metabolomics.

Methods: This study conducted untargeted and targeted metabolomic analyses in a discovery cohort (n = 16) comprising patients with major depressive disorder (MDD) and those who were spontaneously remitted without medication. Findings were validated in an independent cohort (n = 185), comprising drug-treated patients and healthy controls.

Results: Acetylcarnitine levels were significantly increased in spontaneous remission compared to depression, whereas glycerophosphocholine levels were decreased in spontaneous remission. Both metabolites showed the highest concentration in the control group, followed by the remission group, and the lowest concentration in the depression group, regardless of medication status. These changes suggest alterations in mitochondrial and membrane lipid metabolism.

Conclusion: Altered levels in acetylcarnitine and glycerophosphocholine may reflect key pathogenic mechanisms of MDD. These findings offer new insight into spontaneous remission as a distinct clinical subtype of depression and may highlight the potential of these metabolites as biomarkers for treatment monitoring in MDD.

KEY WORDS: Depressive disorder, major; Remission, spontaneous; Metabolomics; Acetylcarnitine; Glycerophosphocholine.

INTRODUCTION

Major depressive disorder (MDD) is a prevalent mental illness characterized by symptoms such as depressed mood, loss of interest, changes in weight or appetite, and

suicidal ideation. It is a major cause of disability worldwide [1,2]. Despite the increased prevalence of MDD and the resulting public health burden [3], the proportion of individuals who recognize the condition and seek (and receive) treatment has not improved [4]. Therefore, fundamental improvements, including community-based efforts, are essential to ensure a better diagnosis and treatment of MDD.

MDD has a multifactorial etiology and exhibits clinical heterogeneity among patients. Therefore, its course and prognosis vary widely [5], presenting a challenge for predicting the effectiveness of each treatment and the selection of the most beneficial therapeutic approach [6,7]. Some patients may achieve remission with treatment, while others may exhibit resistance [8,9] or even sponta-

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neous remission without treatment [10]. Spontaneous remission refers to a reduction of depressive symptoms to subclinical levels without therapeutic intervention [10]. It has been reported that 8 to 18% of untreated patients with depression experience spontaneous remission within 12 weeks [11]. A treatment-free trial in outpatients with depression showed that 32% of patients responded positively during follow-up, suggesting that spontaneous remission is common in clinical settings [12]. This phenomenon can contribute to high remission rates in placebo-treated patients in trials evaluating antidepressant efficacy [13], potentially leading to an overestimation of the short-term benefits of certain antidepressants [12]. Despite the reported impact of spontaneous remission on MDD clinical trials and antidepressant evaluations, its nature and mechanisms are unclear, and are often overlooked in research.

Neurobiological evidence from animal models and human spontaneous remission has provided new insights into the underlying mechanisms of depression, spontaneous remission, and potential pharmacological targets [14-16]. MDD is not only associated with changes in brain structure and function, but also in several physiological systems, such as gastrointestinal, immune, endocrine, neural, hormonal, and oxidative stress pathways [17]. Therefore, to characterize and understand the mechanisms of spontaneous remission in depression, it is necessary to explore biomarkers across multiple biological pathways, along with structural and functional brain data. Metabolites are small molecules produced during metabolic processes and reflect internal physiological changes across different biochemical pathways influenced by the environment, drugs, nutrition, and stress [18]. The metabolic profile of an individual integrates gene expression and protein activity, and metabolites are indicators of interactions between genetic, proteomic, and environmental factors [19]. Metabolomics, which explore the metabolic dynamics of diseases, might be a useful tool to provide insight into the complex pathophysiology of MDD, which involves diverse biological pathways and the interaction between genetic and environmental factors.

This study aimed to explore the molecular profiles of spontaneous remission that differ from those of depressive states using a metabolomics platform. We examined metabolomic biomarkers associated with the clinical characteristics of spontaneous remission and aimed to identify

the molecular mechanisms underlying this phenomenon in depression.

METHODS

Chemicals and Reagents

High-performance liquid chromatography-grade methanol, water, and acetonitrile were purchased from J.T. Baker. Formic acid (mass spectrometry grade) was obtained from Fluka Analytical. Four non-endogenous small molecules (sulfamethoxazole, ketoprofen, MES [4-morpholineethanesulfonic acid], and glyphosate) and 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES), used for spiking sample for quality control of the analysis, were purchased from Sigma-Aldrich. For multiple reaction monitoring (MRM) analysis, standard materials were used to generate calibration curves, and stable isotope-labeled internal standards employed to correct for signal suppression and enhancement [20]. L-acetyl-d3-L-carnitine hydrochloride and glycerophosphocholine standards were purchased from Sigma-Aldrich, Acetylcarnitine from Santa Cruz Biotechnology and Sn-glycero-3-phosphocholine-d9 from Toronto Research Chemicals.

Sample Collection

This study was approved by the Institutional Review Board of Eulji University (IRB No.: EU24-57). The study was conducted in compliance with the Declaration of Helsinki and all participants signed a written informed consent. The patient group consisted of adult outpatients aged 19 years or older, diagnosed with depression by a board-certified psychiatrist according to the Diagnostic and Statistical Manual of Mental Disorders (DSM)-5 criteria. The severity of depression was assessed using the Hamilton Depression Scale (HAM-D)-17 items. The remission status was defined as an asymptomatic period of at least 2 months and HAM-D score of ≤ 7 . The proportion of recurrent MDD patients who experienced two or more depressive episodes was 52.17%. The discovery set included patients with MDD who did not receive medication (ND) and a spontaneous remission group (NR) without medication. The validation set consisted of the ND and NR groups, patients with MDD who received medication (DD), a remission group who received medication (DR), and a healthy control group (Table 1). The DD and DR groups received various types of antidepres-

Table 1. Demographics of participants

Characteristics	Discovery set		Validation set				
	ND	NR	ND	NR	DD	DR	Control
Number of participants	10	6	41	6	39	34	65
Age (yr)	54.60 ± 21.78	54.00 ± 7.56	49.24 ± 19.47	54.00 ± 7.56	50.03 ± 17.02	59.79 ± 15.03	60.20 ± 10.04
Sex (male/female)	3/7	2/4	14/27	2/4	8/31	7/27	12/53
BMI ^a	24.24 ± 4.61	24.96 ± 1.93	24.22 ± 3.53	24.96 ± 1.93	23.61 ± 3.09	24.20 ± 3.26	22.82 ± 4.23
HAMD-17 score	21.60 ± 5.02	0.83 ± 1.33	21.34 ± 3.96	0.83 ± 1.33	20.03 ± 6.49	3.44 ± 3.28	-
Rate of recurrent depression (patients with 2 or more depressive episodes)	40.00%	50.00%	45.00%	50.00%	52.78%	60.61%	-
Drug treatment (yes/no)	No	No	No	No	Yes	Yes	-

Values are presented as number only, %, or mean ± standard deviation.

BMI, body mass index; HAMD-17, Hamilton Depression Rating Scale (17-item); DD, patients with major depressive disorder who received medication; DR, remission group who received medication; ND, patients with major depressive disorder who did not receive medication; NR, the spontaneous remission group who did not receive medication; -, not available.

^aBMI information was not available for three patients in the discovery set and five patients in the validation set.

sants (SSRI, serotonin-norepinephrine reuptake inhibitor [SNRI], tricyclic antidepressant [TCA], and serotonin modulator) either in monotherapy or in combination. Some patients were also prescribed additional medications, including atypical antipsychotics or non-psychiatric drugs such as probiotics (Bioflor) [21], gastrointestinal agents (Gasmotin) [22], circulatory system drugs (Ginexin-F) [23], antiarrhythmics (Indenol) [24], hormonal therapy (Progynova) [25], antihistamines (Twolion) [26], anti-arteriosclerosis drugs (Vytorin) [27]. This cross-sectional study consisted of a cohort of outpatients with MDD, with no strict control over diet, medication, or lifestyle. The control group consisted of adults with no history of brain trauma, psychiatric medication use, and no alcohol or drug abuse in the prior 6 months. Blood samples were collected in a serum tube without anticoagulants, left at room temperature for 30 minutes to allow coagulation, and then centrifuged at 4,000 × g for 5 minutes. The serum was separated, aliquoted into a 1.5-ml microtubes and stored at −70°C until analysis.

Metabolites Extraction and Filtering

Thawed serum (100 µl) was transferred into a 1.5-ml tube and mixed with 200 µl of HPLC grade water by a brief vortex. Subsequently, 700 µl of cold 80% methanol was added, vortexed, and samples were stored at −70°C for quenching. Quenched samples were vortexed for 1 minute, sonicated for 10 minutes, incubated at room temperature for 10 minutes, and centrifuged at 14,000 × g and 4°C for 10 minutes. After centrifugation, 400 µl of the

supernatant was passed through a nanosep filter using a centrifugation step at 14,000 × g and 4°C for 20 minutes. The filtered sample solution was then transferred to a new 1.5-ml tube. This process was repeated, and the filtered sample was dried using a vacuum concentrator (Scan Vac, LaboGene).

Sample Preparation for Untargeted Metabolomics

The dried samples were resuspended in 80 µl of HPLC grade water containing 5% acetonitrile, 0.1% formic acid, and five internal standards (MES, HEPES, and glyphosate at 20 µM concentrations; and ketoprofen and sulfamethoxazole at 10 µM concentrations). Samples were vortexed for 10 minutes, sonicated for 10 minutes, incubated at room temperature for 10 minutes, and centrifuged at 14,000 × g and 4°C for 10 minutes. The supernatant was then diluted 10-fold with the resuspension solution and placed in a mass vial. Analyses were conducted on Agilent 6546 quadrupole time-of-flight (Q-TOF) mass spectrometry system (Agilent Technologies) coupled with Agilent 1290 Infinity II liquid chromatography system. The analytical columns were an Agilent ZORBAX rapid resolution high definition stablebond SB-Aq (2.1 × 150 mm, 1.8 µm) column and an Agilent ZORBAX StableBond-C8 (2.1 × 5 mm, 1.8 µm) guard column, placed in a column oven maintained at 40°C. Mobile phase A consisted of water containing 0.1% formic acid and 5% acetonitrile, and mobile phase B of acetonitrile containing 0.1% formic acid and 5% water. The flow rate of the mobile phase was 400 µl/minutes, and the gradient started at 100% A, de-

creased to 40% over 14 minutes, then decreased by 35% over 3 minutes, maintained at 5% A for 5 minutes, and then returned to 100% A for the final 5 minutes, for a total of 30 minutes. The Vcap voltage of the ion source capillary was set at 4,000 V and the flow rate of the drying nitrogen gas was 11 L/minutes at 225°C. Data were acquired in the positive ion mode at a scanning speed of 2 Hz.

Discovery of Differentially Expressed Metabolites (ND vs. NR)

Raw untargeted mass spectrometry data were imported into Profinder (version 10.0, Agilent Technologies) for molecular features (MF) extraction. The CEF file exported from Profinder was then imported into Mass Profiler Professional (version 15.1, Agilent Technologies) for alignment and statistical analysis using a retention time window of $1\% + 0.15$ minutes and a mass tolerance window of $20\text{ ppm} + 0\text{ mDa}$. Subsequently, MFs commonly detected in all samples of one group were filtered, and volcano plot analysis and partial least squares discriminant analysis (PLS-DA) were performed. In volcano plot, p values were calculated using the Mann-Whitney unpaired test, and MFs satisfying fold change (FC) > 2 and p value < 0.05 were listed by comparing the patients with depression and controls. In PLS-DA, variables with variable importance in projection (VIP) > 1 are generally considered relevant to the model [28], and a higher VIP threshold can be applied to select key variables that improve the specificity of the model [29]. This study used VIP > 2 to select key variables for MFs. MS/MS spectra were then acquired for the selected MFs, and compounds identified using Massbank (<https://massbank.eu/MassBank/>) and the Human Metabolome Database (HMDB, <https://hmdb.ca/>). Metabolite annotation applied levels 2 and 1, as suggested by the Metabolomics standard initiative [30].

Targeted Metabolomics for Metabolite Marker Validation Using Multiple Reaction Monitoring

MRM analysis was performed to validate metabolite markers. Thawed serum (100 μl) was transferred into a 1.5-ml tube and mixed with 100 μl of a 5- $\mu\text{g/ml}$ stable isotope-labeled standard solution. The metabolites were extracted and impurities were filtered, as described for non-targeted analysis sample preparation. Analytical standards were diluted in mobile phase A to prepare nine calibration solutions with concentrations of 0, 5, 10, 50, 100,

150, 200, 250, and 500 ng/ml. MRM analysis was conducted using an Agilent 6495 triple quadrupole (QQQ) system (Agilent Technologies) coupled with an Agilent 1290 Infinity II liquid chromatography system. The columns and mobile phases were the same as those used for non-targeted analysis. The flow rate was 300 $\mu\text{l/minutes}$, and the mobile phase gradient started at 100% A, decreased to 0% over 5 minutes, and was maintained at 100% A from 7 to 10 minutes. Analysis were performed in the positive ion mode with ion source capillary voltage set to 3,000 V, nozzle voltage to 1,500 V, and sheath gas flow rate at 11 L/minutes at 250°C. The target data peaks obtained through mass spectrometry were quantified using MassHunter Quantitative Analysis (version 10.0; Agilent Technologies).

RESULTS

Altered Metabolite Profiles in Untreated Depression and Spontaneous Remission

A total of 2,583 MFs were detected in all samples from at least one group. Volcano plot analysis comparing the ND and NR groups identified 130 MFs that met the criteria of FC > 2.0 and $p < 0.05$. Among these, 92 MFs were upregulated and 38 MFs were downregulated (Fig. 1A). Subsequently, PLS-DA was performed using the 2,538 MF list, yielding a robust distinction model, with an R^2 value of 0.9999 and Q^2 value of 0.8654 (Fig. 1B, 1C). A total of 228 MFs were listed by integrating 130 MFs identified from volcano plot analysis and 170 MFs showing VIP > 2 from PLS-DA (Supplementary Table 1; available online). MS/MS spectra were acquired for these 228 MFs.

Metabolite identification followed the Metabolomics Standard Initiative criteria. For level 2 identification, MS/MS spectra were first matched to experimental data, using Massbank and HMDB. Level 1 identification was confirmed by analyzing standard compounds under identical conditions. Acetylcarnitine and glycerophosphocholine were validated and selected as final markers (Table 2). Acetylcarnitine was significantly increased in the NR group compared with ND ($p = 0.003$, FC = 2.74), while glycerophosphocholine was reduced ($p = 0.001$, FC = 2.72) (Fig. 2).

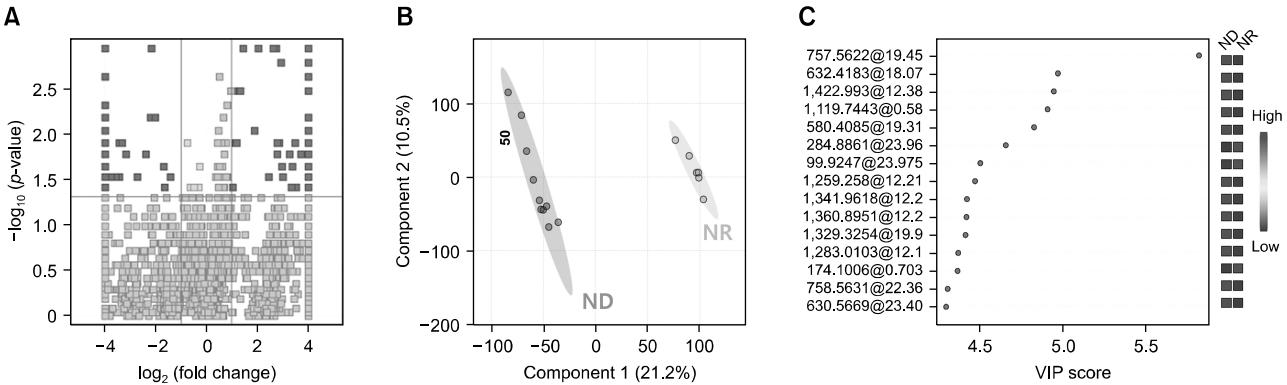


Fig. 1. Metabolic differences between spontaneous remission and depressive states. (A) Volcano plot showing MFs with fold change > 2 and $p < 0.05$ between ND and NR groups. Red squares (top right area divided by lines) represent upregulation, and blue squares (top left area divided by lines) downregulation in ND groups. The horizontal green line represents the corrected p value with the negative common logarithm applied, and the vertical green lines represent the cut-off value of fold change more than twice with the $\log_2$. (B) PLS-DA score plot for comparison of ND vs. NR. (C) Top 15 MFs with VIP > 2.

MF, molecular features; ND, patients with major depressive disorder who did not receive medication; NR, the spontaneous remission group who did not receive medication; PLS-DA, partial least squares discriminant analysis; VIP, variable importance in projection.

Table 2. Compound information of metabolite markers

No.	Compound	Monoisotopic mass	Formula	CAS No.	HMDB ID
1	Acetylcarnitine	203.1158	$C_9H_{17}NO_4$	3040-38-8	HMDB0000201
2	Glycerophosphocholine	257.1028	$C_8H_{20}NO_6P$	28319-77-9	HMDB0000086

HMDB, Human Metabolome Database.

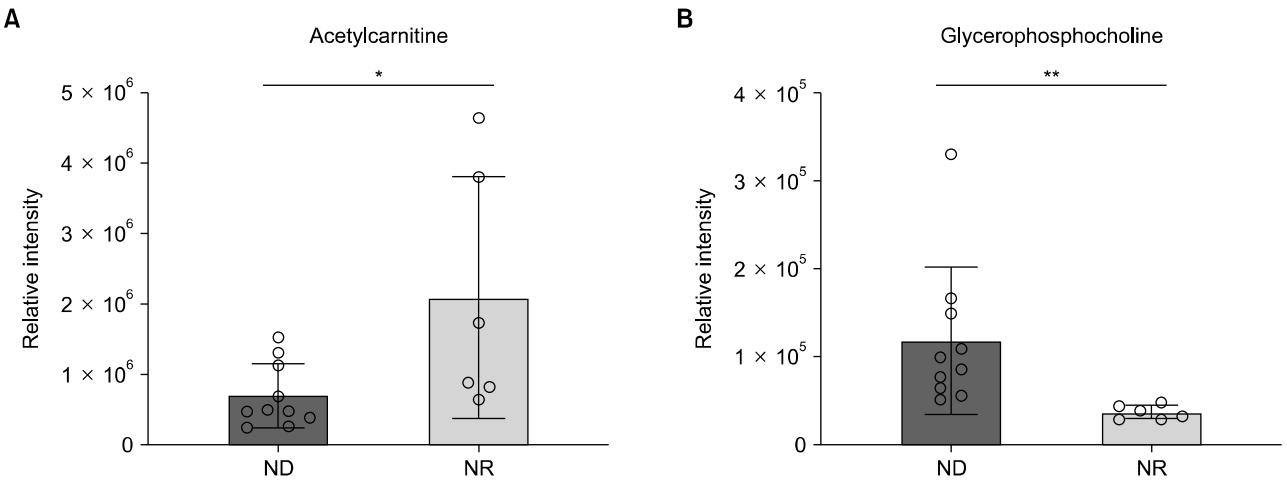


Fig. 2. Relative levels of metabolite markers in spontaneous remission versus depressive states. Scatter plot showing the relative intensity of (A) acetylcarnitine and (B) glycerophosphocholine of ND vs. NR. Each dot represents an individual sample. Data are presented as mean $\pm$ standard deviation. Statistical significance was calculated using the Mann-Whitney unpaired test.

ND, patients with major depressive disorder who did not receive medication; NR, the spontaneous remission group who did not receive medication. $*p < 0.05$, $**p < 0.01$.

Quantitative Validation of Altered Metabolite Markers in Spontaneous Remission

To validate these findings, we examined a larger sam-

ple pool and assessed comparisons according to drug treatment. The validation set included DD, DR, and healthy controls, in addition to ND and NR. Compound opti-

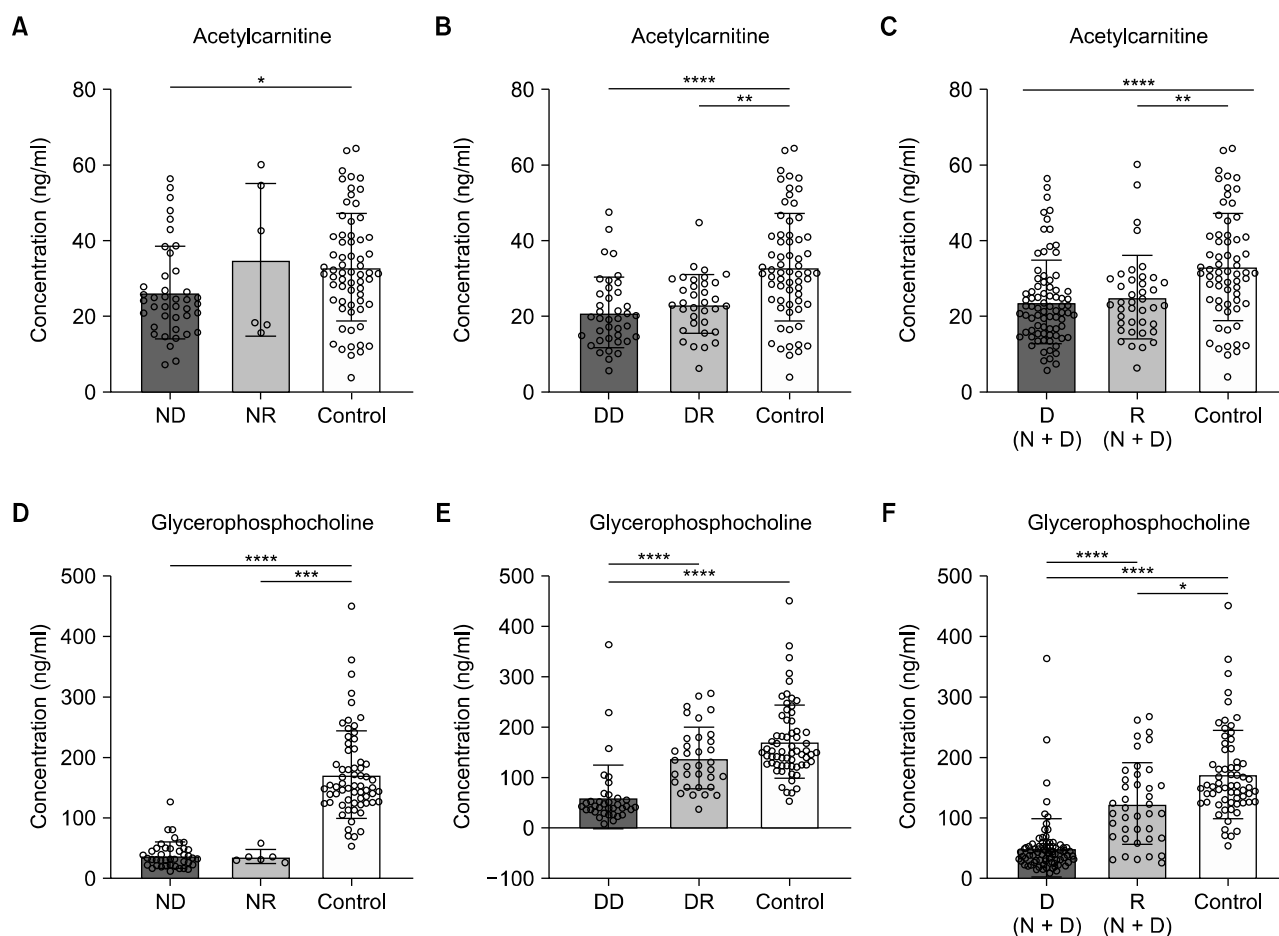


Fig. 3. MRM validation of metabolite markers. Scatter plot showing absolute levels of acetylcarnitine and glycerophosphocholine in ND, NR, DD, DR and healthy control groups. (A) Acetylcarnitine levels in ND, NR, and control groups. (B) Acetylcarnitine levels in DD, DR, and control groups. (C) Acetylcarnitine levels in combined D (ND + DD), combined R (NR + DR), and control groups. (D) Glycerophosphocholine levels in ND, NR, and control groups. (E) Glycerophosphocholine levels in DD, DR, and control groups. (F) Glycerophosphocholine levels in combined D (ND + DD), combined R (NR + DR), and control groups. Each dot represents an individual sample. Data are presented as mean $\pm$ standard deviation. Statistical significance was calculated using the Kruskal-Wallis test.

DD, patients with major depressive disorder who received medication; DR, remission group who received medication; ND, patients with major depressive disorder who did not receive medication; NR, the spontaneous remission group who did not receive medication; D (N + D), all patients with major depressive disorder regardless of medication status (ND + DD); R (N + D), all patients in remission regardless of medication status (NR + DR); MRM, multiple reaction monitoring.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

mization was performed using standard solutions of acetylcarnitine, glycerophosphocholine, and isotope-labeled standard solutions to optimize the MRM transitions of the analytes (Supplementary Table 2; available online).

Similar to the untargeted analysis conducted at the marker discovery stage, the NR group showed an increase in acetylcarnitine and decrease in glycerophosphocholine levels (Fig. 3A, 3D). In the DR group (remission state receiving drug treatment), acetylcarnitine levels were also increased (Fig. 3B), whereas glycerophosphocholine showed the opposite trend, varying with drug treatment (Fig. 3E).

Regardless of drug treatment, acetylcarnitine and glycerophosphocholine showed the highest concentrations in the healthy control group, followed by the remission state, and the lowest concentrations in the depressed state (Fig. 3C, 3F). Glycerophosphocholine showed significant differences in all group comparisons (depression, remission, and healthy) (Fig. 3F).

DISCUSSION

This study identified two serum metabolites signifi-

cantly altered between untreated depression and spontaneous remission. These findings were validated in a large sample pool including healthy controls and individuals receiving medication. Acetylcarnitine consistently increased in remission regardless of medication, while glycerophosphocholine showed the opposite trend depending on medication intervention. Both metabolites increased from depression to remission to healthy controls, with demonstrating differences in all comparisons. These findings may provide new insights into the molecular characteristics and response to therapy of spontaneous remission, one of the clinical subtypes of depression.

Acetylcarnitine, an acetylated derivative of carnitine, is an endogenous compound found in several tissues, including the brain [31]. *De novo* acetylcarnitine synthesis is catalyzed by carnitine acetyltransferase (CAT), located in the inner mitochondrial membrane, endoplasmic reticulum, and peroxide vesicles. CAT catalyzes the transfer of an acetyl group from acetyl-coenzyme A (acetyl-CoA) to carnitine to produce acetylcarnitine and free CoA [32]. The synthesized acetylcarnitine is then transported to the cytosol by the enzyme carnitine/acetylcarnitine translocase (CACT), where it plays a role in the beta-oxidation of fatty acids [33]. Acetylcarnitine promotes the uptake of acetyl-CoA into mitochondria during fatty acid oxidation, enhancing acetylcholine production, stimulating protein and membrane phospholipid synthesis, and preventing excessive neuronal cell death [34]. Previous studies have reported altered acetylcarnitine levels and abnormalities in lipid metabolism in depressive states, potentially resulting in impaired neuroplasticity, a key pathogenic mechanisms of MDD [34]. Furthermore, previous studies have reported decreased acetylcarnitine levels in patients with MDD compared with healthy controls, with normalization in those achieving remission [35,36]. Our findings align with these results, with the decrease in peripheral acetylcarnitine levels in a depressed state being suggestive of impaired mitochondrial metabolism of carnitine to acetylcarnitine, which may ultimately compromise lipid metabolism and mitochondrial energy metabolism in MDD.

As a direct result of intracellular calcium influx and calcium overload, calcium-dependent phospholipase A2 (cPLA2) is activated, and phospholipids, such as phosphatidylcholine and phosphatidylethanolamine, which are major components of cell membranes, are degraded [37]. Phosphatidylcholine and phosphatidylethanolamine are

deacylated to lysophosphatidylcholine and lysophosphatidylethanolamine, respectively, which are then converted to glycerophosphocholine and glycerophosphoethanolamine, respectively, by lysophospholipase [37,38]. A previous study in patients with depression reported that these compounds resulting from membrane phospholipid degradation were increased in the white matter in late-life depression, suggesting that this could result in altered phospholipid metabolism, inflammation, and neurodegeneration [38]. A recent study that integrated functional connectivity (FC) and plasma lipid metabolite profiling of the brains of patients with depression showed that glycerophosphocholine levels were negatively correlated with FC in the right middle frontal gyrus, right anterior cingulate gyrus, and pericingular gyrus, suggesting that changes in FC in patients with MDD are related to synaptic plasticity and serotonin dysregulation [39]. Furthermore, others have reported increased blood glycerophosphocholesterol levels in patients with depression compared to those in controls [40,41], with a reestablishment following treatment with antidepressants [40]. Consistent with these findings, we showed that glycerophosphocholine levels increased in untreated depression compared with that in the spontaneous remission state. Integrated analysis of the gut microbiota profile and blood metabolites suggested that the oxidation of glycerophosphocholine may contribute to the pathogenesis of MDD, through the NOD-like receptor family pyrin domain containing 3 (NLRP3) inflammasome pathway and lipid metabolism [42]. Therefore, the observed changes in glycerophosphocholine levels may provide a link to the inflammatory and lipid metabolism abnormalities observed in MDD.

Additionally, animal studies suggest that acetylcarnitine administration alters the composition of the mitochondrial inner membrane proteins in the cerebellum and affects the levels of glycerophosphocholine, potentially through the interaction with cytoskeletal proteins and alteration of membrane dynamics in the vicinity of the glycerol backbone of the phospholipid bilayer, contributing to membrane stability [33]. Together, these two metabolites may reflect changes in membrane lipid metabolism and mitochondrial energy metabolism, associated with the pathophysiological mechanisms of depression. Both acetylcarnitine and glycerophosphocholine can cross the blood-brain barrier, supporting their relevance as peripheral biomarkers of central changes in MDD. The admin-

istration of these metabolites in animals promoted antidepressant, anti-inflammatory, and neuroprotective effects [43,44], suggesting that changes in the endogenous levels of these metabolites might be related to pathogenic mechanisms of depression, and be indicative of therapeutic responses, highlighting their potential as biomarkers for MDD.

This study shows some limitations. First, the main treatment strategy for MDD is antidepressants, as such, it is difficult to recruit patients that show spontaneous remission without drug treatment. Therefore, the sample size of the spontaneous remission group was relatively small compared with that of the other groups, which may limit statistical power. Validation in a large sample group that considers appropriate effect size and power is necessary to strengthen our findings. Among all MDD patients, the proportion of recurrent patients with 2 or more depressive episodes was 52.17%. Previous studies have reported changes in the proteomic profile in relapse [45], and this heterogeneity of depressive states may also affect the metabolomic profiles. Although sex and body mass index were matched between groups, age differences in some groups within the validation may have introduced variability (Supplementary Table 3; available online). In addition, a subset of MDD patients presented comorbidities (assessed through a questionnaire), including diabetes (15.35%), hypertension (27.83%), hyperlipidemia (20.87%), and gastrointestinal diseases (7.83%). These diseases involve metabolic changes, which can represent confounding variables. This study is a preliminary study based on a cohort of outpatient MDD patients, and rigorous control over lifestyle, medication, and comorbidities was not feasible. Future research should include subgroup analyses to account for these variables, which will allow a more refined exploration of metabolic dynamics related to MDD. Furthermore, although volcano plot analysis was used in the discovery phase, multiple testing correction was not applied. Considering the large number of metabolites examined, it is necessary to control the influence of type 1 error through multiple testing. In addition, quantitative changes in the overall associated metabolic processes were not identified. Future work should explore network-level alterations and underlying mechanism to establish comprehensive metabolic key signatures and pathways in MDD. These analyses would enable the identification of reliable biomarkers and novel therapeutic targets.

In conclusion, we identified altered metabolic profiles in spontaneous remission compared with those in untreated depression. The changes in acetylcarnitine and glycerophosphocholine may be related to the pathophysiology of MDD, reflecting dysregulated lipid metabolism related to mitochondria and cell membranes. These findings provide new information regarding spontaneous remission, a subtype of depression whose molecular mechanisms are poorly understood, and present promising biomarkers for the subtyping and treatment of MDD.

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■ Conflicts of Interest

No potential conflict of interest relevant to this article was reported.

■ Author Contributions

Conceptualization: Seungyeon Lee, Hee-Gyoo Kang, Jiyeong Lee. Formal analysis: Seungyeon Lee, Sora Mun, Yeeun Yun, Myoung Soo Woo. Investigation: Hee-Gyoo Kang, Jiyeong Lee. Methodology: Seungyeon Lee, Sora Mun, Yeeun Yun, Myoung Soo Woo. Supervision: Hee-Gyoo Kang, Jiyeong Lee. Visualization: Seungyeon Lee. Writing—original draft: Seungyeon Lee. Writing—review & editing: Seungyeon Lee, Hee-Gyoo Kang, Jiyeong Lee.

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REFERENCES

1. Belmaker RH, Agam G. *Major depressive disorder*. *N Engl J Med* 2008;358:55-68.
2. Cui L, Li S, Wang S, Wu X, Liu Y, Yu W, et al. *Major depressive disorder: hypothesis, mechanism, prevention and treatment*. *Signal Transduct Target Ther* 2024;9:30.
3. Yan G, Zhang Y, Wang S, Yan Y, Liu M, Tian M, et al. *Global, regional, and national temporal trend in burden of major de-*

- pressive disorder from 1990 to 2019: an analysis of the global burden of disease study. *Psychiatry Res* 2024;337:115958.
4. Greenberg PE, Fournier AA, Sisitsky T, Simes M, Berman R, Koenigsberg SH, et al. The economic burden of adults with major depressive disorder in the United States (2010 and 2018). *Pharmacoeconomics* 2021;39:653-665.
 5. Malhi GS, Mann JJ. Depression. *Lancet* 2018;392:2299-2312.
 6. Cuijpers P, Stringaris A, Wolpert M. Treatment outcomes for depression: challenges and opportunities. *Lancet Psychiatry* 2020;7:925-927.
 7. Serretti A. A critical view on new and future antidepressants. *Clin Psychopharmacol Neurosci* 2024;22:201-210.
 8. Kim J, Kim TE, Lee SH, Koo JW. The role of glutamate underlying treatment-resistant depression. *Clin Psychopharmacol Neurosci* 2023;21:429-446.
 9. Anmella G, Meehan A, Ashton M, Mohebbi M, Fico G, Ng CH, et al. Exploring clinical subgroups of participants with major depressive disorder that may benefit from adjunctive minocycline treatment. *Clin Psychopharmacol Neurosci* 2024;22:33-44.
 10. Whiteford HA, Harris MG, McKeon G, Baxter A, Pennell C, Barendregt JJ, et al. Estimating remission from untreated major depression: a systematic review and meta-analysis. *Psychol Med* 2013;43:1569-1585.
 11. Mekonen T, Ford S, Chan GCK, Hides L, Connor JP, Leung J. What is the short-term remission rate for people with untreated depression? A systematic review and meta-analysis. *J Affect Disord* 2022;296:17-25.
 12. Posternak MA, Zimmerman M. Short-term spontaneous improvement rates in depressed outpatients. *J Nerv Ment Dis* 2000;188:799-804.
 13. Ormel J, Kessler RC, Schoevers R. Depression: more treatment but no drop in prevalence: how effective is treatment? And can we do better? *Curr Opin Psychiatry* 2019;32:348-354.
 14. Khalid A, Kim BS, Seo BA, Lee ST, Jung KH, Chu K, et al. Gamma oscillation in functional brain networks is involved in the spontaneous remission of depressive behavior induced by chronic restraint stress in mice. *BMC Neurosci* 2016;17:4.
 15. Moraga-Amaro R, Vazquez-Matias DA, Nazario LR, Dierckx RAJO, Stehberg J, Doorduyn J, et al. Increased dopamine D₂/D₃ receptor and serotonin transporter availability in male rats after spontaneous remission from repeated social defeat-induced depression; a PET study in rats. *Neurobiol Dis* 2024;202:106727.
 16. Amone D, McKie S, Elliott R, Juhasz G, Thomas EJ, Downey D, et al. State-dependent changes in hippocampal grey matter in depression. *Mol Psychiatry* 2013;18:1265-1272.
 17. Stolfi F, Abreu H, Sinella R, Nembrini S, Centonze S, Landra V, et al. Omics approaches open new horizons in major depressive disorder: from biomarkers to precision medicine. *Front Psychiatry* 2024;15:1422939.
 18. Nedic Erjavec G, Konjevod M, Nikolac Perkovic M, Svob Strac D, Tudor L, Barbas C, et al. Short overview on metabolic approach and redox changes in psychiatric disorders. *Redox Biol* 2018;14:178-186.
 19. Quinones MP, Kaddurah-Daouk R. Metabolomics tools for identifying biomarkers for neuropsychiatric diseases. *Neurobiol Dis* 2009;35:165-176.
 20. Hewavitharana AK. Matrix matching in liquid chromatography-mass spectrometry with stable isotope labelled internal standards—is it necessary? *J Chromatogr A* 2011;1218:359-361.
 21. Heo J, Kim SK, Park KS, Jung HK, Kwon JG, Jang BI. A double-blind, randomized, active drug comparative, parallel-group, multi-center clinical study to evaluate the safety and efficacy of probiotics (*Bacillus licheniformis*, Zhengchangsheng® capsule) in patients with diarrhea. *Intest Res* 2014;12:236-244.
 22. Park JH, Lee KN, Lee OY, Choi MG, Chung H, Choi SC, et al. Efficacy and safety of DWJ1252 compared with Gasmotin in the treatment of functional dyspepsia: a multicenter, randomized, double-blind, active-controlled study. *J Neurogastroenterol Motil* 2021;27:87-96.
 23. Hong YH, Mun SK. A case of sudden unilateral sensorineural hearing loss with contralateral psychogenic hearing loss induced by gunshot noise. *Mil Med* 2011;176:1193-1195.
 24. Choi M, Jung II, Lee BY, Kim HR. Ventricular septal aneurysms in adults: findings of cardiac CT images and correlation with clinical features. *Acta Radiol* 2011;52:619-623.
 25. Song Z, Song K, Zhao H, He Y, Hu J. Network analysis and experimental approach to investigate the potential therapeutic mechanism of zishen yutai pills on premature ovarian insufficiency. *Heliyon* 2023;9:e20025.
 26. Kim YB, Choi HS, Cho HK, Seo GW. Diagnosis and treatment of bullous pemphigoid that developed twice after total knee replacement arthroplasty: a case report. *BMC Musculoskelet Disord* 2021;22:118.
 27. Meaney A, Ceballos G, Asbun J, Solache G, Mendoza E, Vela A, et al. The VYtorin on Carotid intima-media thickness and overall arterial rigidity (VYCTOR) study. *J Clin Pharmacol* 2009;49:838-847.
 28. Akarachantachote N, Chadcham S, Saithanu K. Cutoff threshold of variable importance in projection for variable selection. *Int J Pure Appl Math* 2014;94:307-322.
 29. Troisi J, Colucci A, Cavallo P, Richards S, Symes S, Landolfi A, et al. A serum metabolomic signature for the detection and grading of bladder cancer. *Appl Sci (Basel)* 2021;11:2835.
 30. Sumner LW, Amberg A, Barrett D, Beale MH, Beger R, Daykin CA, et al. Proposed minimum reporting standards for chemical analysis Chemical Analysis Working Group (CAWG) Metabolomics Standards Initiative (MSI). *Metabolomics* 2007;3:211-221.
 31. Jones LL, McDonald DA, Borum PR. Acylcarnitines: role in brain. *Prog Lipid Res* 2010;49:61-75.
 32. Chiechio S, Canonico PL, Grilli M. L-acetylcarnitine: a mechanistically distinctive and potentially rapid-acting antidepressant drug. *Int J Mol Sci* 2017;19:11.

33. Pettegrew JW, Levine J, McClure RJ. *Acetyl-L-carnitine physical-chemical, metabolic, and therapeutic properties: relevance for its mode of action in Alzheimer's disease and geriatric depression. Mol Psychiatry* 2000;5:616-632.
34. Wang SM, Han C, Lee SJ, Patkar AA, Masand PS, Pae CU. *A review of current evidence for acetyl-L-carnitine in the treatment of depression. J Psychiatr Res* 2014;53:30-37.
35. Lee S, Mun S, Lee YR, Choi H, Joo EJ, Kang HG, et al. *Discovery and validation of acetyl-L-carnitine in serum for diagnosis of major depressive disorder and remission status through metabolomic approach. Front Psychiatry* 2022;13:1002828.
36. Ait Tayeb AEK, Colle R, El-Asmar K, Chappell K, Acquaviva-Bourdain C, David DJ, et al. *Plasma acetyl-L-carnitine and L-carnitine in major depressive episodes: a case-control study before and after treatment. Psychol Med* 2023;53:2307-2316.
37. Walter A, Korth U, Hilgert M, Hartmann J, Weichel O, Hilgert M, et al. *Glycerophosphocholine is elevated in cerebrospinal fluid of Alzheimer patients. Neurobiol Aging* 2004;25:1299-1303.
38. Harper DG, Jensen JE, Ravichandran C, Sivrioglu Y, Silveri M, Iosifescu DV, et al. *Tissue-specific differences in brain phosphodiesterases in late-life major depression. Am J Geriatr Psychiatry* 2014;22:499-509.
39. Wei J, Zhang Z, Yang X, Zhao L, Wang M, Dou Y, et al. *Abnormal functional connectivity within the prefrontal cortex is associated with multiple plasma lipid species in major depressive disorder. J Affect Disord* 2024;350:713-720.
40. Umehara H, Numata S, Watanabe SY, Hatakeyama Y, Kinoshita M, Tomioka Y, et al. *Altered KYN/TRP, Gln/Glu, and Met/methionine sulfoxide ratios in the blood plasma of medication-free patients with major depressive disorder. Sci Rep* 2017;7:4855.
41. Kawamura N, Shinoda K, Sato H, Sasaki K, Suzuki M, Yamaki K, et al. *Plasma metabolome analysis of patients with major depressive disorder. Psychiatry Clin Neurosci* 2018;72:349-361.
42. Zhao H, Jin K, Jiang C, Pan F, Wu J, Luan H, et al. *A pilot exploration of multi-omics research of gut microbiome in major depressive disorders. Transl Psychiatry* 2022;12:8.
43. Nasca C, Xenos D, Barone Y, Caruso A, Scaccianoce S, Matrisciano F, et al. *L-acetylcarnitine causes rapid antidepressant effects through the epigenetic induction of mGlu2 receptors. Proc Natl Acad Sci U S A* 2013;110:4804-4809.
44. Umino M, Okuda M, Ohkubo T, Fujii T, Matsubara K. *Long-term intake of α -glycerophosphocholine (GPC) suppresses microglial inflammation and blood-brain barrier (BBB) disruption and promotes neurogenesis in senescence-accelerated mice prone 8 (SAMP8). Biosci Biotechnol Biochem* 2023;87:1537-1542.
45. Yun Y, Mun S, Lee S, Kang HG, Lee J. *Serum L-selectin levels as predictive markers for chronic major depressive disorder progression. Ann Gen Psychiatry* 2024;23:37.